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In [1]: # Program 14: Decision Tree classifier on Iris
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

FILE = r"D:\Datascience-MachineLearning\Datasets\iris_dataset.csv" # <-- change to your file path

df = pd.read_csv(FILE)
if "Id" in df.columns:
    df = df.drop(columns="Id")
X = df.iloc[:, :-1] # features
y = df.iloc[:, -1] # last column = species

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

print("Accuracy:", accuracy_score(y_test, model.predict(X_test)))

# predict classes of new test samples (sepal length/width, petal length/width)
new_samples = pd.DataFrame([[5.1, 3.5, 1.4, 0.2],
                             [6.7, 3.0, 5.2, 2.3]], columns=X.columns)
print("Predictions for new samples:", model.predict(new_samples))
```

Accuracy: 1.0

Predictions for new samples: ['Iris-setosa' 'Iris-virginica']

In []: